SAT Math Mastery · Session 4

Equivalent Expressions and Nonlinear Equations

Time 3 hours: Part A in class (about 2 hours 20 minutes of solving), then Part B, the self-assessment, done independently (about 45 minutes), and a short error log.

Topics quadratic equations: discriminant, sum and product of solutions · absolute-value and rational equations · equivalent rational expressions · exponent rules and radicals · factoring and polynomial division · matching coefficients · undefined values

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. HARD · DESMOS

For nonzero real numbers p and q, the equation $px^2 + 3qx + 4p = 0$ has exactly one real solution, and $pq = -48$. If $p > 0$, what is the value of $p + q$?

(A) -14 **(B) -2 ✓**

(C) 2 (D) 14

1. Exactly one real solution means the discriminant $(3q)^2 - 4(p)(4p)$ is 0. Put $q = -\dfrac{48}{p}$ (from $pq = -48$) and let x stand for p: type `y=9(48/x)^2-16x^2`.

2. Tap the zero with $x > 0$ (we need $p > 0$): (6, 0). So $p = 6$.

3. Type `-48/6`: it returns -8, so $q = -8$.

4. Type `6+(-8)`: it returns -2. Option (B).

Answer: (B)

2. HARD · DESMOS

The equation $x^2 - 9x + k = 0$, where k is a constant, has two real solutions that differ by 3. What is the value of k?

1. For $x^2 - 9x + k = 0$ the two solutions differ by $\sqrt{81 - 4k}$. Let x stand for k: type `y=sqrt(81-4x)` and `y=3`.

2. Tap the intersection: (18, 3). So $k = 18$.

3. Check: type `y=x^2-9x+18` and tap its zeros (3, 0) and (6, 0).

4. Type `6-3`: it returns 3: the solutions differ by 3.

Answer: 18

3. HARD · DESMOS

The equation $2x^2 - 10x + c = 0$, where c is a constant, has a product of solutions equal to 6. What is the sum of the reciprocals of the two solutions?

(A) $\frac{1}{5}$ (B) $\frac{1}{30}$

(C) $\frac{5}{6}$ ✓ (D) $\frac{6}{5}$

1. Type `6*2`: it returns 12: the product of the solutions is $\frac{c}{2} = 6$, so $c = 12$.
2. Type `y=2x^2-10x+12` and tap the zeros (2, 0) and (3, 0).
3. Type `1/2+1/3`: it returns 0.8333, which is $\frac{5}{6}$ (the sum of the reciprocals).
4. Check with sum over product: type `5/6`: it returns 0.8333 (the sum of solutions is $\frac{10}{2} = 5$ and the product is 6). Option (C).

Answer: (C)

4. HARD · DESMOS

Let r and s be the two solutions of $x^2 - 7x + 3 = 0$. What is the value of $r^2 + s^2$?

1. Type `y=x^2-7x+3`.
2. Tap the two zeros: (0.4586, 0) and (6.5414, 0). These are r and s.
3. For exact values (discriminant $49 - 12 = 37$): type `(7-sqrt(37))/2`: it returns 0.4586 and type `(7+sqrt(37))/2`: it returns 6.5414.
4. Type `((7-sqrt(37))/2)^2+((7+sqrt(37))/2)^2`: it returns 43.

Answer: 43

5. HARD · DESMOS

The equation $x^2 - 5x + 6 = 0$ has solutions p and q. Which of the following equations has solutions $2p$ and $2q$?

(A) $x^2 - 10x + 24 = 0$ ✓ (B) $x^2 - 10x + 12 = 0$

(C) $x^2 - 5x + 12 = 0$ (D) $2x^2 - 5x + 6 = 0$

1. Type `y=x^2-5x+6` and tap the zeros (2, 0) and (3, 0): $p = 2$, $q = 3$.
2. Doubling them: type `2*2`: it returns 4 and type `2*3`: it returns 6.
3. Type the options as graphs. `y=x^2-10x+24` has its zeros at (4, 0) and (6, 0), which is what we want.
4. A wrong option such as `y=x^2-10x+12` has zeros (1.3944, 0) and (8.6056, 0) instead. Option (A).

Answer: (A)

6. HARD · DESMOS

The equation $x^2 - bx + 27 = 0$, where b is a positive constant, has two real solutions, and one solution is 3 times the other. What is the value of b?

1. Call the solutions r and $3r$; their product is $3r^2 = 27$. Let x stand for r: type `y=3x^2` and `y=27`.
2. Tap the intersection with $x > 0$: (3, 27). So $r = 3$.
3. Type `3+3*3`: it returns 12, so $b = r + 3r = 12$.
4. Check: type `y=x^2-12x+27` and tap the zeros (3, 0) and (9, 0) ($9 = 3 \cdot 3$).

Answer: 12

7. HARD · DESMOS

$$x^2 - 6x = 2|x - 3|$$

What is the sum of all real solutions of the equation above?

(A) 9

(B) 0

(C) 3

(D) 6 ✓

1. Type `y=x^2-6x` and `y=2abs(x-3)`.
2. Tap both intersections: (−1.1623, 8.3246) and (7.1623, 8.3246).
3. Type `-1.1623+7.1623`: it returns 6, so the sum of the solutions is 6.
4. Why: the parabola and the V both have their axis at $x = 3$, so the two points are the same distance from 3 and their x-values add to $2 \cdot 3$. Option (D).

Answer: (D)

8. HARD · DESMOS

$$\frac{x}{x-4} = 2 + \frac{4}{x-4}$$

How many solutions does the equation above have?

(A) Zero ✓

(B) Exactly one

(C) Exactly two

(D) Infinitely many

1. Type `y=x/(x-4)` and `y=2+4/(x-4)`.
2. The curves never touch. The second is always above the first: type `2+4/(5-4)`: it returns 6 and type `5/(5-4)`: it returns 5 at $x = 5$, and type `2+4/(8-4)`: it returns 3 and type `8/(8-4)`: it returns 2 at $x = 8$.
3. Algebra agrees. Multiply by $x - 4$: $x = 2(x - 4) + 4$, so $x = 2x - 4$ and $x = 4$.
4. But $x = 4$ makes the denominator 0, so it is not allowed. The equation has no solution. Option (A).

Answer: (A)

9. HARD · DESMOS

The function f is defined by $f(x) = ax^2 + bx + c$, where a, b, and c are constants. The graph of $y = f(x)$ in the xy-plane passes through $(1, 8)$ and $(-1, 2)$ and has exactly one x-intercept. What is the greatest possible value of a?

(A) $\frac{5}{2}$ (B) 9

(C) $\frac{9}{2}$ ✓ (D) $\frac{1}{2}$

1. Type `(8-2)/2`: it returns 3 (subtracting $f(1) = 8$ and $f(-1) = 2$ gives $2b$), so $b = 3$ and $c = 5 - a$.
2. One x-intercept means the discriminant is 0. Let x stand for a: type `y=3^2-4x(5-x)`.
3. Tap the zeros: (0.5, 0) and (4.5, 0). So $a = \frac{1}{2}$ or $a = \frac{9}{2}$.
4. The greatest is $a = 4.5$. Check: type `3^2-4*4.5*0.5`: it returns 0, so the graph of `y=4.5x^2+3x+0.5` only touches the x-axis. Option (C).

Answer: (C)

10. HARD · DESMOS

The expression $\frac{3}{x-4} - \frac{2}{x+1}$ is equivalent to which of the following, for $x \neq 4$ and $x \neq -1$?

(A) $\frac{x+11}{x^2+3x-4}$ **(B) $\frac{x+11}{x^2-3x-4}$ ✓**

(C) $\frac{x-11}{x^2-3x-4}$ (D) $\frac{5x-5}{x^2-3x-4}$

1. Type the expression from the question as `f(x)=3/(x-4)-2/(x+1)`.
2. Type the options as their own functions. Only `g(x)=(x+11)/(x^2-3x-4)` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=(x-11)/(x^2-3x-4)` leaves it.
3. Check at $x = 2$: type `f(2)`: it returns -2.1667; type `g(2)`: it returns -2.1667 (same); type `h(2)`: it returns 1.5 (different).
4. Algebra behind it: the numerator is $3(x+1) - 2(x-4) = x + 11$ over $(x-4)(x+1) = x^2 - 3x - 4$. Option (B).

Answer: (B)

11. HARD · DESMOS

The expression $\frac{5}{x+2} + \frac{3x}{x^2+x-2}$ is equivalent to which of the following, for $x \neq -2$ and $x \neq 1$?

(A) $\frac{8x-5}{x^2-x-2}$ (B) $\frac{3x+5}{x^2+x-2}$

(C) $\frac{8x+5}{x^2+x-2}$ **(D) $\frac{8x-5}{x^2+x-2}$ ✓**

1. Type the expression from the question as `f(x)=5/(x+2)+3x/(x^2+x-2)`.
2. Type the options as their own functions. Only `g(x)=(8x-5)/(x^2+x-2)` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=(3x+5)/(x^2+x-2)` leaves it.
3. Check at $x = 3$: type `f(3)`: it returns 1.9; type `g(3)`: it returns 1.9 (same); type `h(3)`: it returns 1.4 (different).
4. Algebra behind it: $x^2 + x - 2 = (x+2)(x-1)$, so $\frac{5}{x+2} = \frac{5x-5}{x^2+x-2}$ and the numerators add to $8x - 5$. Option (D).

Answer: (D)

12. HARD · DESMOS

The operation $\odot$ is defined for all real numbers $x \neq 1$ by $x^{\odot} = \dfrac{x+3}{x-1}$. Which of the following is equivalent to $x^{\odot} - 1$?

(A) $\dfrac{4}{x-1}$ ✓

(B) $\dfrac{2x+4}{x-1}$

(C) $\dfrac{2}{x-1}$

(D) $\dfrac{x+2}{x-1}$

1. Type the expression from the question as `f(x)=(x+3)/(x-1)-1`.
2. Type the options as their own functions. Only `g(x)=4/(x-1)` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=(x+2)/(x-1)` leaves it.
3. Check at $x = 3$: type `f(3)`: it returns 2; type `g(3)`: it returns 2 (same); type `h(3)`: it returns 2.5 (different).
4. Algebra behind it: $(x+3) - (x-1) = 4$. Option (A).

Answer: (A)

13. HARD · DESMOS

If $a(x-2)(x+6) = a(x-h)^2 + k$ for all values of x, where $a \neq 0$, what is k in terms of a?

(A) $16a$

(B) $-16a$ ✓

(C) $-12a$

(D) $4a$

1. Type `f(x)=a(x-2)(x+6)` (accept the slider for a) and set $a = 3$.
2. Tap the vertex: $(-2, -48)$. Its y-value is k for $a = 3$.
3. Type `-48/3`: it returns -16, so $k = -16a$ (and the vertex is at $h = -2$).
4. Check with $a = -2$: type `-2*(-2-2)*(-2+6)`: it returns 32 is the vertex height, and type `-16*(-2)`: it returns 32 agrees. Option (B).

Answer: (B)

14. HARD · DESMOS

Which expression is equivalent to $x^4 - 2x^2 - 15$?

(A) $\left(x^2 - 15\right)\left(x^2 + 1\right)$

(B) $\left(x^2 - 1\right)\left(x^2 + 15\right)$

(C) $\left(x^2 - 5\right)\left(x^2 + 3\right)$ ✓

(D) $\left(x^2 + 5\right)\left(x^2 - 3\right)$

1. Type the expression from the question as `f(x)=x^4-2x^2-15`.
2. Type the options as their own functions. Only `g(x)=(x^2-5)(x^2+3)` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=(x^2+5)(x^2-3)` leaves it.
3. Check at $x = 2$: type `f(2)`: it returns -7; type `g(2)`: it returns -7 (same); type `h(2)`: it returns 9 (different).
4. Algebra behind it: with $u = x^2$, $u^2 - 2u - 15 = (u-5)(u+3)$. Option (C).

Answer: (C)

15. HARD · DESMOS

If $\frac{x^{a^2}}{x^{b^2}} = x^{20}$, $x > 1$, and $a + b = 4$, what is the value of $a - b$?

1. Equal powers of $x > 1$ have equal exponents, so $a^2 - b^2 = 20$, and $a^2 - b^2 = (a + b)(a - b) = 4(a - b)$. Let x stand for $a - b$: type `y=4x` and `y=20`.
2. Tap the intersection: (5, 20). So $a - b = 5$.
3. Check: type `(4+5)/2`: it returns 4.5 is a, and type `(4-5)/2`: it returns -0.5 is b.
4. Type `4.5^2-(-0.5)^2`: it returns 20, as required.

Answer: 5

16. HARD · NO CALCULATOR NEEDED

If $x > 0$ and $\sqrt{x} = x\sqrt{x} = x^2 = \sqrt{x^2}$, what is the value of x?

1. Type `y=sqrt(x)` and `y=x*sqrt(x)`.
2. Tap the intersections: (0, 0) and (1, 1). Only $x = 1$ has $x > 0$.
3. Check the other two expressions at $x = 1$: type `1^2`: it returns 1 and type `sqrt(1^2)`: it returns 1.
4. All four expressions equal 1 at $x = 1$, so $x = 1$.

Answer: 1

17. HARD · DESMOS

If $s^2 = t^{-1}$ and $t^{1/3} = u^{-1/2}$, then what is the value of s in terms of u, for $s > 0$?

(A) $u^{3/4}$ ✓ (B) $u^{-3/4}$

(C) $u^{-1/2}$ (D) $u^{1/2}$

1. Test with $u = 16$. From $t^{1/3} = u^{-1/2}$: type `(16^(-1/2))^3`: it returns 0.0156, which is t.
2. Then $s^2 = \frac{1}{t}$: type `sqrt(1/((16^(-1/2))^3))`: it returns 8, so $s = 8$.
3. Try the options at $u = 16$: type `16^(3/4)`: it returns 8 matches, while type `16^(-3/4)`: it returns 0.125 does not.
4. Also type `16^(1/2)`: it returns 4 is not 8. So $s = u^{3/4}$. Option (A).

Answer: (A)

18. HARD · DESMOS

Let $\langle x \rangle$ be defined by $\langle x \rangle = \dfrac{3x+1}{x-2}$ for any x such that $x \neq 2$. Which of the following is equivalent to $\langle x \rangle - 3$?

(A) $\dfrac{6x-5}{x-2}$

(B) $\dfrac{7}{x-2} - 3$

(C) $\dfrac{3x-2}{x-2}$

(D) $\dfrac{7}{x-2}$ ✓

1. Type the expression from the question as `f(x)=(3x+1)/(x-2)-3`.
2. Type the options as their own functions. Only `g(x)=7/(x-2)` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=(6x-5)/(x-2)` leaves it.
3. Check at $x = 3$: type `f(3)`: it returns 7; type `g(3)`: it returns 7 (same); type `h(3)`: it returns 13 (different).
4. Algebra behind it: $(3x+1) - 3(x-2) = 7$. Option (D).

Answer: (D)

19. HARD · DESMOS

Let a and b be numbers such that $a^4 = b^3$. Which of the following is equivalent to $a\sqrt{b}$?

(A) $b^{4/3}$

(B) $b^{3/2}$

(C) $b^{5/4}$ ✓

(D) $b^{7/6}$

1. Test with $b = 16$. From $a^4 = b^3$: type `(16^3)^(1/4)`: it returns 8, so $a = 8$.
2. Then $a\sqrt{b}$: type `8*sqrt(16)`: it returns 32.
3. Try the options at $b = 16$: type `16^(5/4)`: it returns 32 matches.
4. A wrong option such as type `16^(4/3)`: it returns 40.3175 does not. Option (C).

Answer: (C)

20. HARD · DESMOS

What is the value of $\left(\dfrac{3x}{y}\right)\left(\dfrac{x^{-1}}{(3y)^{-1}}\right)$?

(A) 6

(B) 9 ✓

(C) 1

(D) 3

1. Type the expression with $x = 2$ and $y = 5$: type `(3*2/5)*(2^(-1)/(3*5)^(-1))`: it returns 9.
2. Try other values, $x = 4$ and $y = 7$: type `(3*4/7)*(4^(-1)/(3*7)^(-1))`: it returns 9.
3. The value does not depend on x or y; in general type `(3*2/5)*(3*5/2)`: it returns 9 (the negative exponents flip to $\dfrac{3y}{x}$).
4. The value is always 9. Option (B).

Answer: (B)

21. HARD · NO CALCULATOR NEEDED

Find the value of $\dfrac{\sqrt{c+7}}{c} \cdot c\sqrt{c}$ if $c = 9$.

1. Type `c=9`.

2. Type the expression: type `sqrt(c+7)/c*c*sqrt(c)` : it returns 12.

3. Check by cancelling c: type `sqrt(c+7)*sqrt(c)` : it returns 12.

4. By hand: type `sqrt(16)*3` : it returns 12.

Answer: 12

22. HARD · DESMOS

Evaluate $\left(\dfrac{5}{a^2} + \dfrac{4}{3a}\right)^a$ for $a = -3$.

1. Type `a=-3`.

2. Type the bracket: type `5/a^2+4/(3a)` : it returns 0.1111, which is $\dfrac{1}{9}$.

3. Type the whole expression: type `(5/a^2+4/(3a))^a` : it returns 729.

4. Check by hand: type `9^3` : it returns 729 (a negative exponent flips $\dfrac{1}{9}$ to 9).

Answer: 729

23. HARD · DESMOS

For $x \neq -3$ and $x \neq 2$, the expression $\dfrac{x^2 - 2x - 15}{x^2 + x - 6}$ is equivalent to $\dfrac{x+a}{x+b}$, where a and b are constants. What is the value of $a - b$?

1. Type `f(x)=(x^2-2x-15)/(x^2+x-6)` and `g(x)=(x-5)/(x-2)`.

2. The two graphs lie on top of each other (apart from the hole at $x = -3$), so $a = -5$ and $b = -2$.

3. Check at $x = 4$: type `f(4)` : it returns -0.5 and type `g(4)` : it returns -0.5.

4. Type `-5-(-2)` : it returns -3.

Answer: -3

24. HARD · DESMOS

If $a^{\frac{3}{4}} = 5$ and $a > 0$, what is the value of $a^{-\frac{3}{2}}$?

1. Solve for a: type `5^(4/3)` : it returns 8.5499. Type `a=5^(4/3)`.

2. Check the condition: type `a^(3/4)` : it returns 5.

3. Now the asked value: type `a^(-3/2)` : it returns 0.04.

4. As a fraction: type `1/25` : it returns 0.04, so the answer is $\dfrac{1}{25}$.

Answer: $\dfrac{1}{25}$ **(or** 0.04**)**

25. HARD · DESMOS

If $a^2 + b^2 = 4ab$, where $a \neq b$ and $b \neq 0$, what is the value of $\dfrac{a+b}{a-b}$?

(A) 2 (B) 3

(C) 6 **(D) $\pm\sqrt{3}$ ✓**

1. Take $b = 1$: the equation becomes $a^2 + 1 = 4a$. Type `y=x^2-4x+1`.
2. Tap the zeros: (0.2679, 0) and (3.7321, 0), which are $a = 2 \pm \sqrt{3}$.
3. Evaluate $\dfrac{a+b}{a-b}$ for each: type `(2+sqrt(3)+1)/(2+sqrt(3)-1)`: it returns 1.7321, and type `(2-sqrt(3)+1)/(2-sqrt(3)-1)`: it returns -1.7321.
4. Compare type `sqrt(3)`: it returns 1.7321: the values are $\pm\sqrt{3}$. Option (D).

Answer: (D)

26. HARD · DESMOS

The equation $\dfrac{ax+b}{x+2} = 3 - \dfrac{11}{x+2}$ is true for all values of x not equal to -2, where a and b are constants. What is the value of ab?

1. Type `f(x)=3-11/(x+2)`.
2. Type `g(x)=(ax+b)/(x+2)` (accept the sliders for a and b) and slide until g lies exactly on f: $a = 3$, $b = -5$.
3. Check at $x = 1$: type `3-11/(1+2)`: it returns -0.6667 and type `(a*1+b)/(1+2)`: it returns -0.6667.
4. Type `a*b`: it returns -15.

Answer: -15

27. HARD · DESMOS

The expression $10x^4 + 33x^2 + 20$ can be rewritten as $(5x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(5x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + d$?

1. Expand $(5x^2 + c)(2x^2 + d) = 10x^4 + (5d + 2c)x^2 + cd$. Matching gives $cd = 20$ and $5d + 2c = 33$ (the same for a, b).
2. Put $d = \dfrac{20}{c}$ into $5d + 2c = 33$ and multiply by c: $100 + 2c^2 = 33c$, so $2c^2 - 33c + 100 = 0$. Type `y=2x^2-33x+100` (letting x stand for c).
3. Tap the zeros: (4, 0) and (12.5, 0). The whole-number zero 4 is a (with $b = 5$). The other zero is $c = 12.5$, and then type `20/12.5`: it returns 1.6, so $d = 1.6$ is a positive nonintegral value.
4. Type `4+1.6`: it returns 5.6, which is $\dfrac{28}{5}$.

Answer: $\dfrac{28}{5}$ **(or** 5.6**)**

28. HARD · DESMOS

$$\frac{40x+100}{2x^2-5x-25}+\frac{x^2y-5xy}{4xy}$$

Which of the following is equivalent to the expression above for $x>5$, $y>0$?

(A) $\dfrac{x^2-10x+105}{4(x-5)}$ ✓

(B) $\dfrac{x^2-10x+105}{4(2x+5)}$

(C) 5

(D) $\dfrac{x-1}{x+15}$

1. The letter y cancels in the second fraction (since $y>0$), so test with $y=2$. Type `f(x)=(40x+100)/(2x^2-5x-25)+(2x^2-10x)/(8x)` and look at $x>5$.
2. Type each option as its own graph, for example `g(x)=(x^2-10x+105)/(4(x-5))` and `h(x)=(x^2-10x+105)/(4(2x+5))`. Graph g lies exactly on f; graph h does not.
3. Confirm at $x=7$: type `f(7)`: it returns 10.5; type `g(7)`: it returns 10.5 (same); type `h(7)`: it returns 1.1053 (different). Option (A).
4. Algebra: $\dfrac{40x+100}{2x^2-5x-25}=\dfrac{20(2x+5)}{(2x+5)(x-5)}=\dfrac{20}{x-5}$ and the second fraction is $\dfrac{x-5}{4}$. Over $4(x-5)$ the numerator is $80+(x-5)^2=x^2-10x+105$.

Answer: (A)

29. HARD · DESMOS

For real numbers p and q with $p^2+4q^2=72$, the product $(2x+p)(x+q)$ is equivalent to $2x^2+bx+c$, where b and c are constants. What is the greatest possible value of c?

1. Expand $(2x+p)(x+q)=2x^2+(2q+p)x+pq$, so $c=pq$. With $q=\sqrt{\dfrac{72-p^2}{4}}$ and x standing for p, type `y=x*sqrt((72-x^2)/4)`.
2. Tap the highest point: (6, 18). So the greatest c is 18, at $p=6$.
3. Check: type `sqrt((72-6^2)/4)`: it returns 3 is q, and type `6^2+4*3^2`: it returns 72 confirms $p^2+4q^2=72$.
4. Type `6*3`: it returns 18.

Answer: 18

30. HARD · DESMOS

For constants p, q and c, the equation

$$(2x-3)(x^2+px+q)=2x^3-5x^2+cx+9$$

is true for all real values of x. What is the value of c?

1. Matching the x^2 terms gives $2p-3=-5$: type `(-5+3)/2`: it returns -1, so $p=-1$. Matching constants gives $-3q=9$: type `9/(-3)`: it returns -3, so $q=-3$.
2. The x terms give $c=2q-3p$: type `2*(-3)-3*(-1)`: it returns -3.
3. Check: type `y=(2x-3)(x^2-x-3)` and `y=2x^3-5x^2-3x+9`; the graphs coincide.
4. At $x=2$: type `f(2)`: it returns -1 and type `g(2)`: it returns -1.

Answer: -3

31. HARD · DESMOS

For a positive real number x, $\dfrac{\sqrt[3]{x}\cdot\sqrt[4]{x^3}}{\sqrt[6]{x}}=\sqrt[3]{k}$, where $k=2^{11}$. Which of the following is equal to x?

(A) $2^{\frac{44}{3}}$

(B) 2^4 ✓

(C) $2^{\frac{11}{3}}$

(D) $2^{\frac{121}{36}}$

1. Type `y=x^(1/3)*x^(3/4)/x^(1/6)` and `y=(2^11)^(1/3)` ($k^{1/3}=(2^{11})^{1/3}$).
2. Tap the intersection: (16, 12.6992). So $x=16$.
3. Check: type `16^(1/3)*16^(3/4)/16^(1/6)` : it returns 12.6992 and type `(2^11)^(1/3)` : it returns 12.6992.
4. Type `2^4` : it returns 16. Option (B).

Answer: (B)

32. HARD · DESMOS

For $x>0$ and any positive integer n, which of the following is equivalent to $\dfrac{\left(x^{2n+1}\right)^3+4x^{6n+3}-2x\left(x^{3n+1}\right)^2}{x^{6n+1}}$?

(A) $3x^4$

(B) $3x^{12n+4}$

(C) $3x^2$ ✓

(D) $5x^2$

1. Test with numbers: type `x=2` and `n=1`.
2. Type the expression: type `((x^(2n+1))^3+4x^(6n+3)-2x(x^(3n+1))^2)/x^(6n+1)` : it returns 12.
3. Try the options: type `3x^2` : it returns 12 matches, while type `5x^2` : it returns 20 and type `3x^4` : it returns 48 do not.
4. Repeat with $n=2$ (type `n=2`): type `3x^2` : it returns 12 is still the value of the expression: type `((x^(2n+1))^3+4x^(6n+3)-2x(x^(3n+1))^2)/x^(6n+1)` : it returns 12. Option (C).

Answer: (C)

33. HARD · DESMOS

For real numbers $k\neq 3$ and $x\neq 0$:

$$\frac{4(k-3)}{x(k-3)}+\frac{2}{x}-\frac{3}{4}=\frac{3}{8}$$

What is the value of x?

1. Type `k=7` (any $k\neq 3$ works), then `y=4(k-3)/(x(k-3))+2/x-3/4` and `y=3/8`.
2. Tap the intersection: (5.3333, 0.375).
3. As a fraction: type `16/3` : it returns 5.3333, so $x=\dfrac{16}{3}$.
4. Check with another k, say $k=-2$: type `4(-2-3)/((16/3)(-2-3))+2/(16/3)-3/4` : it returns 0.375.

Answer: $\dfrac{16}{3}$

34. HARD · DESMOS

The function f is defined by $f(x) = \dfrac{9^x \cdot 12^{x+1}}{8^x}$. Which of the following expressions is equivalent to $f(x)$ and displays the value of $f(2)$ as a coefficient?

(A) $2187\left(\dfrac{27}{2}\right)^{x-2}$ ✓

(B) $2187\left(\dfrac{27}{2}\right)^{x+2}$

(C) $162\left(\dfrac{27}{2}\right)^{x-1}$

(D) $\dfrac{729}{4}\left(\dfrac{27}{2}\right)^{x-2}$

1. Type `f(x)=9^x*12^(x+1)/8^x` and `g(x)=2187(27/2)^(x-2)`.
2. The graphs coincide. Check: type `f(2)`: it returns 2187 and type `g(3)`: it returns 29524.5 (the same as type `f(3)`: it returns 29524.5).
3. So g shows $f(2) = 2187$ as its coefficient (exponent $x - 2$).
4. A wrong option such as `h(x)=729/4*(27/2)^(x-2)` gives type `h(2)`: it returns 182.25, not $f(2)$. Also type `f(1)`: it returns 162 is the coefficient of $162\left(\frac{27}{2}\right)^{x-1}$, which shows $f(1)$. Option (A).

Answer: (A)

35. HARD · DESMOS

For integers a, b, c, and d, $(ax + b)(cx + d) = 6x^2 + kx + 5$ for all values of x, where k is a constant. Which of the following must be true?

I. k is odd
II. When k is divided by 4, the remainder is 1
III. k is not divisible by 5

(A) II and III only

(B) I, II, and III

(C) I only

(D) I and III only ✓

1. From $ac = 6$ and $bd = 5$ take $b = 1$, $d = 5$, so $k = ad + bc = 5a + c$. Type the factor pairs as lists: `a=[1,2,3,6]` and `c=6/a`.
2. Type `5a+c`: it returns 11, 13, 17, 31. (Other choices of b, d give these or their negatives.)
3. Remainders: type `mod(5a+c,4)`: it returns 3, 1, 1, 3 (not always 1, so II is false) and type `mod(5a+c,5)`: it returns 1, 3, 2, 1 (never 0, so III is true).
4. Every value is odd: type `mod(5a+c,2)`: it returns 1, 1, 1, 1, so I is true. Statements I and III only. Option (D).

Answer: (D)

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. MEDIUM · NO CALCULATOR NEEDED

Which of the following is equivalent to $\dfrac{x^2-9}{x-3}$, for $x \neq 3$?

(A) x^2-3 (B) $x+9$

(C) $x+3$ ✓ (D) $x-3$

1. Type the expression from the question as `f(x)=(x^2-9)/(x-3)`.
2. Type the options as their own functions. Only `g(x)=x+3` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=x-3` leaves it.
3. Check at $x=5$: type `f(5)`: it returns 8; type `g(5)`: it returns 8 (same); type `h(5)`: it returns 2 (different).
4. Algebra behind it: $x^2-9=(x-3)(x+3)$ and $(x-3)$ cancels. Option (C).

Answer: (C)

37. MEDIUM · NO CALCULATOR NEEDED

Which of the following is equivalent to $\dfrac{x^2+x-12}{x+4}$, for $x \neq -4$?

(A) x^2-3 **(B) $x-3$ ✓**

(C) $x+3$ (D) $x-4$

1. Type the expression from the question as `f(x)=(x^2+x-12)/(x+4)`.
2. Type the options as their own functions. Only `g(x)=x-3` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=x+3` leaves it.
3. Check at $x=2$: type `f(2)`: it returns -1; type `g(2)`: it returns -1 (same); type `h(2)`: it returns 5 (different).
4. Algebra behind it: $x^2+x-12=(x+4)(x-3)$ and $(x+4)$ cancels. Option (B).

Answer: (B)

38. MEDIUM · DESMOS

If a and b are positive numbers such that $a^5=b^2$, which of the following is equivalent to $b\sqrt{a}$?

(A) $b^{9/4}$ (B) $b^{1/5}$

(C) $b^{9/5}$ **(D) $b^{6/5}$ ✓**

1. Test with $b=32$. From $a^5=b^2$: type `(32^2)^(1/5)`: it returns 4, so $a=4$.
2. Then $b\sqrt{a}$: type `32*sqrt(4)`: it returns 64.
3. Try the options at $b=32$: type `32^(6/5)`: it returns 64 matches, while type `32^(9/5)`: it returns 512 does not.
4. Also type `32^(9/4)`: it returns 2435.4962 is far too large. Option (D).

Answer: (D)

39. Medium · Desmos

What is the quotient when $2x^3 - 3x^2 - 11x + 6$ is divided by $x - 3$ (the division has no remainder)?

(A) $2x^2 + 3x - 2$ ✓ (B) $2x^2 - 3x - 2$

(C) $2x^2 + 3x + 2$ (D) $2x^2 - 9x + 16$

1. Type the expression from the question as `f(x)=(2x^3-3x^2-11x+6)/(x-3)`.
2. Type the options as their own functions. Only `g(x)=2x^2+3x-2` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=2x^2-3x-2` leaves it.
3. Check at $x = 4$: type `f(4)`: it returns 42; type `g(4)`: it returns 42 (same); type `h(4)`: it returns 18 (different).
4. Algebra behind it: synthetic division by 3 leaves quotient $2x^2 + 3x - 2$ and remainder 0. Option (A).

Answer: (A)

40. Medium · No calculator needed

Which expression is equivalent to $4(2x + 7) - 5x$?

(A) $13x + 28$ **(B)** $3x + 28$ ✓

(C) $3x + 2$ (D) $13x + 7$

1. Type the expression from the question as `f(x)=4(2x+7)-5x`.
2. Type the options as their own functions. Only `g(x)=3x+28` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=13x+28` leaves it.
3. Check at $x = 2$: type `f(2)`: it returns 34; type `g(2)`: it returns 34 (same); type `h(2)`: it returns 54 (different).
4. Algebra behind it: $8x + 28 - 5x = 3x + 28$. Option (B).

Answer: (B)

41. Medium · No calculator needed

Which expression is equivalent to $(5x^2 + 3x - 9) - (2x^2 - 7x + 4)$?

(A) $7x^2 - 4x - 5$ (B) $7x^2 + 10x - 5$

(C) $3x^2 + 10x - 13$ ✓ (D) $3x^2 - 4x - 13$

1. Type the expression from the question as `f(x)=(5x^2+3x-9)-(2x^2-7x+4)`.
2. Type the options as their own functions. Only `g(x)=3x^2+10x-13` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=3x^2-4x-13` leaves it.
3. Check at $x = 2$: type `f(2)`: it returns 19; type `g(2)`: it returns 19 (same); type `h(2)`: it returns -9 (different).
4. Algebra behind it: subtracting the bracket changes every sign inside it. Option (C).

Answer: (C)

42. Medium · No calculator needed

An urban planner models the area of a rectangular section of a park as $3x^2 + ax$ square meters, where a is a constant. This expression is equivalent to $x\,(3x + 11)$. What is the value of a?

(A) 11 ✓ (B) 14

(C) 3 (D) 8

1. Type `f(x)=3x^2+ax` (accept the slider for a) and `g(x)=x(3x+11)`.
2. Slide a until f lies exactly on g: $a = 11$.
3. Check at $x = 2$: type `3*2^2+a*2`: it returns 34 and type `2*(3*2+11)`: it returns 34.
4. Also at $x = -1$: type `3*(-1)^2+a*(-1)`: it returns -8 and type `-1*(3*(-1)+11)`: it returns -8. Option (A).

Answer: (A)

43. Medium · Desmos

The sum of $-3x^2 + 5x + 18$ and $6x^2 - 2x - 9$ can be written in the form $ax^2 + bx + c$, where a, b, and c are constants. What is the value of $a + b + c$?

1. Type `f(x)=(-3x^2+5x+18)+(6x^2-2x-9)`.
2. The value $a + b + c$ is $f(1)$: type `f(1)`: it returns 15.
3. Check by adding coefficients: type `-3+6`: it returns 3 is a, type `5-2`: it returns 3 is b, type `18-9`: it returns 9 is c.
4. Type `3+3+9`: it returns 15.

Answer: 15

44. Medium · No calculator needed

Which of the following is equivalent to $\dfrac{2x^2 + 8x}{2x}$, for $x \neq 0$?

(A) $x + 8$ (B) $2x + 4$

(C) $x^2 + 4$ **(D)** $x + 4$ ✓

1. Type the expression from the question as `f(x)=(2x^2+8x)/(2x)`.
2. Type the options as their own functions. Only `g(x)=x+4` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=x+8` leaves it.
3. Check at $x = 3$: type `f(3)`: it returns 7; type `g(3)`: it returns 7 (same); type `h(3)`: it returns 11 (different).
4. Algebra behind it: $2x^2 + 8x = 2x(x + 4)$ and $2x$ cancels. Option (D).

Answer: (D)

45. Medium · Desmos

If $x > 0$ and $\sqrt{x\sqrt{x}} = x^a$, what is the value of a?

1. Test with $x = 16$: type `sqrt(16*sqrt(16))`: it returns 8.
2. Type `y=16^x` and `y=8` (x stands for a).
3. Tap the intersection: (0.75, 8). So $a = 0.75$.
4. Exact: type `ln(8)/ln(16)`: it returns 0.75, which is $\dfrac{3}{4}$.

Answer: $\dfrac{3}{4}$

46. Medium · Desmos

If $a^{b/5} = 32$ for positive integers a and b, where a is a prime number, what is the value of b?

1. $32 = 2^5$ and a is prime, so $a = 2$. Type `y=2^(x/5)` and `y=32` (x stands for b).
2. Tap the intersection: (25, 32). So $b = 25$.
3. Check: type `2^(25/5)` : it returns 32.
4. Another prime fails: type `3^(25/5)` : it returns 243 is not 32.

Answer: 25

47. Medium · Desmos

Which of the following is a factor of the polynomial $(x^2 - 9)(x + 7)$?

(A) $x + 5$ (B) $x - 2$

(C) $x + 3$ ✓ (D) $x - 4$

1. Type `f(x)=(x^2-9)(x+7)` .
2. Tap the zeros: $(-7, 0)$, $(-3, 0)$ and $(3, 0)$. A zero at $x = r$ means $x - r$ is a factor.
3. So the factors are $x + 7$, $x + 3$ and $x - 3$. Check the zero -3: type `f(-3)` : it returns 0.
4. The other options' zeros are not zeros of f: type `f(2)` : it returns -45, type `f(4)` : it returns 77, type `f(-5)` : it returns 32. Option (C).

Answer: (C)

48. Medium · No calculator needed

If $k = m^2n^2$, k is not 0, and the values of m and n are both tripled, then the value of k increases by which of the following factors?

(A) 27 **(B) 81 ✓**

(C) 3 (D) 9

1. Test with numbers: type `m=2` and `n=5`, then type `m^2n^2` : it returns 100.
2. Triple both: type `(3m)^2(3n)^2` : it returns 8100.
3. The factor: type `8100/100` : it returns 81.
4. Try other values, $m = 1$ and $n = 4$: type `(3*1)^2*(3*4)^2/(1^2*4^2)` : it returns 81. Option (B).

Answer: (B)

49. Medium · Desmos

In the equation $\sqrt{200} + \sqrt{200} = m\sqrt{n}$, m and n are integers such that $m > n$ and n is not divisible by the square of any integer greater than 1. What is the value of m?

1. Type the left side: type `sqrt(200)+sqrt(200)` : it returns 28.2843.
2. Try $m = 20$, $n = 2$: type `20*sqrt(2)` : it returns 28.2843 (the same value).
3. The other pair with $m > n$ is $m = 10$, $n = 8$: type `10*sqrt(8)` : it returns 28.2843, but 8 is divisible by 4, so it is not allowed.
4. Check squares: type `20^2*2` : it returns 800 and type `(2*sqrt(200))^2` : it returns 800. So $m = 20$.

Answer: 20

50. Medium · Desmos

The value of y decreased by 5 is directly proportional to the value of x increased by 3. If $y = 7$ when $x = 1$, what is the value of x when $y = 19$?

1. Write $y - 5 = k(x + 3)$. With $y = 7$, $x = 1$: type `(7-5)/(1+3)` : it returns 0.5, so $k = 0.5$.

2. Type `y=0.5(x+3)+5` and `y=19` .

3. Tap the intersection: (25, 19). So $x = 25$.

4. Check: type `0.5*(25+3)+5` : it returns 19.

Answer: 25

51. Medium · Desmos

The expression $\dfrac{11p^2 + 11q^2}{9p^2 + 30p + 9}$ is undefined when

(A) $p = 0$ or $q = 0$

(B) $q = -\dfrac{1}{3}$ or $q = -3$

(C) $p = -3$ or $q = -\dfrac{1}{3}$

(D) $p = -\dfrac{1}{3}$ or $p = -3$ ✓

1. A fraction is undefined when its denominator is 0. The numerator does not matter, and the denominator has only p in it, so q cannot make it undefined.

2. Type `y=9x^2+30x+9` (letting x stand for p) and tap the zeros.

3. The zeros are $(-3, 0)$ and $(-0.3333, 0)$, so $p = -3$ or $p = -\dfrac{1}{3}$.

4. Check by factoring: $9p^2 + 30p + 9 = 3(3p + 1)(p + 3)$, and $3(3p + 1)(p + 3) = 9x^2 + 30x + 9$. Option (D).

Answer: (D)

52. Medium · Desmos

For $x > 0$, which expression is equivalent to $\sqrt{(9x^2)^3 \cdot x^6}$?

(A) $27x^6$ ✓

(B) $27x^9$

(C) $81x^6$

(D) $3x^6$

1. Type the expression from the question as `f(x)=sqrt((9x^2)^3*x^6)` .

2. Type the options as their own functions. Only `g(x)=27x^6` lies exactly on the graph of f (one curve); a wrong option such as `h(x)=81x^6` leaves it.

3. Check at $x = 1.5$: type `f(1.5)` : it returns 307.5469; type `g(1.5)` : it returns 307.5469 (same); type `h(1.5)` : it returns 922.6406 (different).

4. Algebra behind it: $(9x^2)^3 x^6 = 729x^{12}$ and $\sqrt{729x^{12}} = 27x^6$. Option (A).

Answer: (A)

53. Medium · Desmos

If $\left(\sqrt{p}\right)^6 = q^{-3}$ and $q = \dfrac{1}{4}$, what is the value of p?

1. Type `y=sqrt(x)^6` and `y=(1/4)^(-3)` ($q^{-3} = \left(\dfrac{1}{4}\right)^{-3} = 64$), with x standing for p.

2. Tap the intersection: (4, 64). So $p = 4$.

3. Check: type `sqrt(4)^6` : it returns 64 and type `(1/4)^(-3)` : it returns 64.

4. Also $\left(\sqrt{p}\right)^6 = p^3$: type `4^3` : it returns 64.

Answer: 4

54. Medium · Desmos

$\dfrac{\sqrt[3]{x^7}}{\sqrt{x}} = x^{a/b}$ for $x > 1$, where a and b are positive integers with no common factor. What is the value of $\dfrac{a}{b}$?

1. Type `f(x)=(x^7)^(1/3)/sqrt(x)` and `g(x)=x^(11/6)`.

2. The graphs coincide. Check at $x = 2$: type `f(2)`: it returns 3.5636 and type `g(2)`: it returns 3.5636.

3. The exponent: type `7/3-1/2`: it returns 1.8333, which is $\dfrac{11}{6}$.

4. Compare type `11/6`: it returns 1.8333. The fraction $\dfrac{11}{6}$ has no common factor.

Answer: $\dfrac{11}{6}$

55. Medium · No calculator needed

If $x + \dfrac{1}{x} = 5$, what is the value of $x^2 + \dfrac{1}{x^2}$?

1. Type `y=x+1/x` and `y=5`.

2. Tap an intersection: (4.7913, 5).

3. Exact value (discriminant $25 - 4 = 21$): type `(5+sqrt(21))/2`: it returns 4.7913.

4. Type `((5+sqrt(21))/2)^2+1/((5+sqrt(21))/2)^2`: it returns 23.

Answer: 23

56. Medium · Desmos

$$\frac{2}{x-2} + \frac{3}{x+5} = \frac{rx+t}{(x-2)(x+5)}$$

The equation above is true for all $x \neq 2, -5$, where r and t are constants. What is the value of rt?

(A) 15 **(B) 20 ✓**

(C) 60 (D) -20

1. Type `f(x)=2/(x-2)+3/(x+5)` and `g(x)=(5x+4)/((x-2)(x+5))` ($r = 5$, $t = 4$ after combining over $(x-2)(x+5)$).

2. The graphs coincide. Check at $x = 1$: type `f(1)`: it returns -1.5 and type `g(1)`: it returns -1.5.

3. The numerator: type `2*(1+5)+3*(1-2)`: it returns 9 at $x = 1$, which equals $5(1) + 4$: type `5*1+4`: it returns 9.

4. Type `5*4`: it returns 20. Option (B).

Answer: (B)

57. Medium · Desmos

$\dfrac{\left(\sqrt[3]{x^2}\right)^5}{x^2} = x^{a/b}$ for $x > 1$, where a and b are positive integers with no common factor. What is the value of $\dfrac{a}{b}$?

1. Type `f(x)=((x^2)^(1/3))^5/x^2` and `g(x)=x^(4/3)`.

2. The graphs coincide. Check at $x = 8$: type `f(8)`: it returns 16 and type `g(8)`: it returns 16.

3. The exponent: type `2/3*5-2`: it returns 1.3333, which is $\dfrac{4}{3}$.

4. Compare type `4/3`: it returns 1.3333. The fraction $\dfrac{4}{3}$ has no common factor.

Answer: $\dfrac{4}{3}$